Fundamentals of Electrical and Electronics Engineering (FEEE)

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T-P-C: 3-2-4

Lecture-22

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AC Power Analysis

1. Instantaneous Power
2. Average Power/ Effective Power / Real Power/Active Power
3. Apparent Power
4. Complex Power(Real Power + j Reactive Power)

1. Instantaneous power (in watts)

$$p(t) = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2}V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

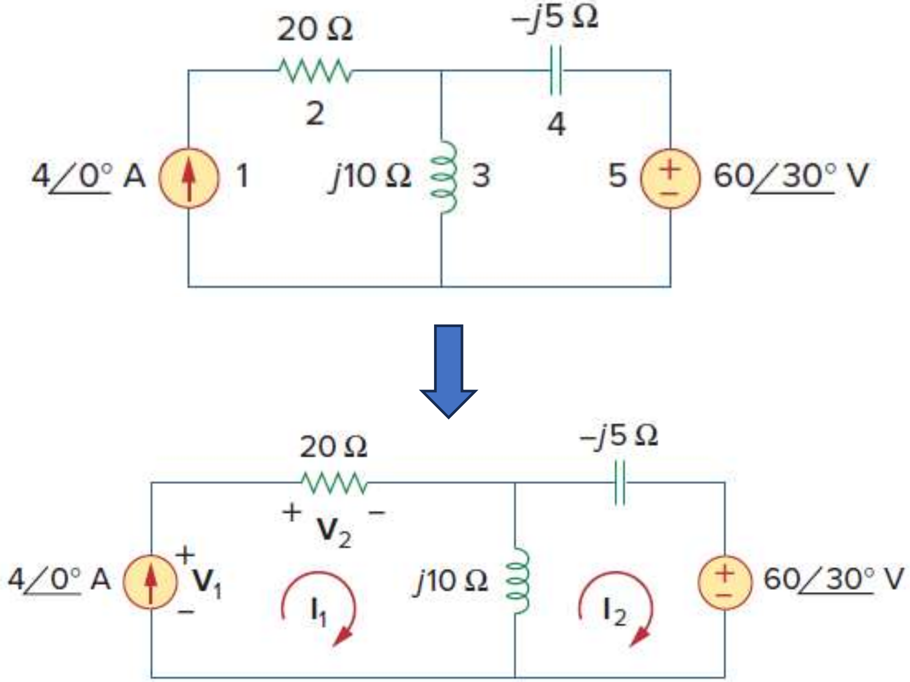
2. Average Power (in watts)

$$P = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i)$$

$$= \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}}\cos(\theta_v - \theta_i) \quad = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

= Effective Power / Real Power/Active Power

6. Determine the average power generated by each source and the average power absorbed by each passive element in the circuit of Fig. below.



We apply mesh analysis as shown in Fig. (b). For mesh 1,

$$\mathbf{I}_1 = 4 \text{ A}$$

For mesh 2,

$$(j10 - j5)\mathbf{I}_2 - j10\mathbf{I}_1 + 60\underline{/30^\circ} = 0,$$

$$j5\mathbf{I}_2 = -60\underline{/30^\circ} + j40$$

$$\mathbf{I}_2 = -12\underline{/-60^\circ} + 8$$
$$= 10.58\underline{/79.1^\circ} \text{ A}$$

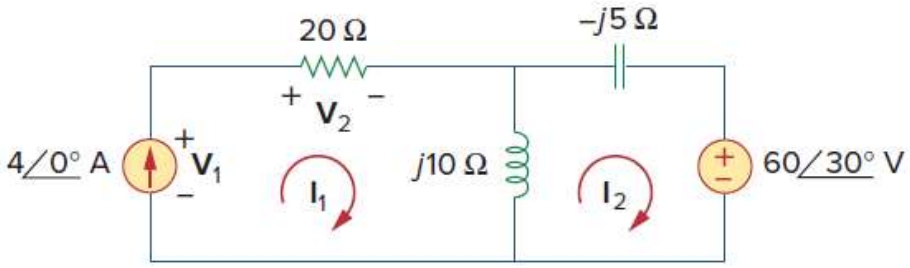
the voltage across it is $60\underline{/30^\circ}$ V, so that the average power is

$$P_5 = \frac{1}{2}(60)(10.58)\,\text{os}(30^\circ - 79.1^\circ) = 207.8 \text{ W}$$

(1)-Power absorbed

For the resistor, the current through it is $\mathbf{I}_1 = 4\underline{/0^\circ}$ and the voltage across it is $20\mathbf{I}_1 = 80\underline{/0^\circ}$, so that the power absorbed by the resistor is

$$P_2 = \frac{1}{2}\ (80)(4) = 160\ \text{W} \qquad (2)$$



For the current source, the current through it is $\mathbf{I}_1 = 4\underline{/0^\circ}$ and the voltage across it is

$$\mathbf{V}_1 = 20\mathbf{I}_1 + j10(\mathbf{I}_1 - \mathbf{I}_2)$$

$$= 80 + j10(4 - 2 - j10.39)$$

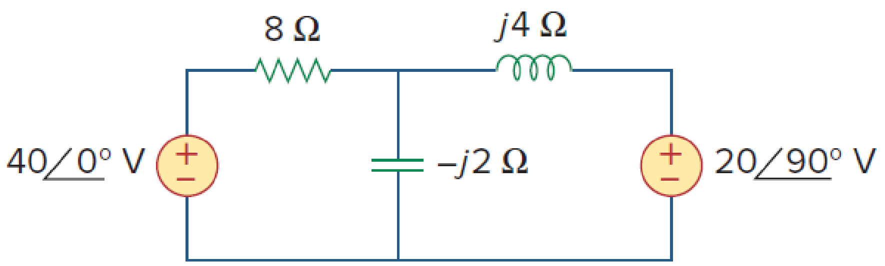
$$= 183.9 + j20 = 184.984\underline{/6.21^\circ}\ \text{V}$$

(1) + (2) = (3)

The average power supplied by the current source is

$$P_1 = \frac{1}{2}\ (184.984)(4)\ \text{os}(6.21^\circ - 0) = 367.8\ \text{W} \qquad (3)$$

7. Calculate the average power of the five elements in the circuit of Fig. below.



Answer: 40-V Voltage source: 60 W; *j*20-V Voltage source: 40 W; resistor: 100 W; others: 0 W.

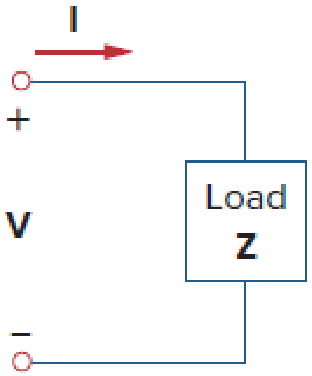
8).

A current $\mathbf{I} = 33\underline{/30^\circ}$ A flows through an impedance $\mathbf{Z} = 40\underline{/-22^\circ}$ Ω. Find the average power delivered to the impedance.

3) Complex Power

in phasor form, $\mathbf{V} = V_m\angle\theta_v$ and $\mathbf{I} = I_m\angle\theta_i$,

$$\mathbf{S} = I_{\text{rms}}^2\,\mathbf{Z} = \frac{V_{\text{rms}}^2}{\mathbf{Z}^*} = \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}}^*$$



- the *complex power* **S** absorbed by the ac load is

$$\mathbf{S} = \frac{1}{2}\mathbf{V}\mathbf{I}^* \quad \text{VA} \qquad \mathbf{V}_{\text{rms}} = \frac{\mathbf{V}}{\sqrt{2}} = V_{\text{rms}}\angle\theta_v$$

$$\mathbf{S} = \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}}^* \qquad \mathbf{I}_{\text{rms}} = \frac{\mathbf{I}}{\sqrt{2}} = I_{\text{rms}}\angle\theta_i$$

$$\mathbf{S} = V_{\text{rms}}I_{\text{rms}}\angle\theta_v - \theta_i$$

$$= V_{\text{rms}}I_{\text{rms}}\cos(\theta_v - \theta_i) + jV_{\text{rms}}I_{\text{rms}}\sin(\theta_v - \theta_i)$$

$$= P + jQ$$

Real Power $\quad P = V_{\text{rms}}I_{\text{rms}}\cos(\theta_v - \theta_i)$ W

Reactive Power $\quad Q = V_{\text{rms}}I_{\text{rms}}\sin(\theta_v - \theta_i)$ VAR

4) Apparent Power $= S = |\mathbf{S}| = |\mathbf{V}_{\text{rms}}|\,|\mathbf{I}_{\text{rms}}| = \sqrt{P^2 + Q^2}$

4). Apparent Power and Power Factor

$$v(t) = V_m \cos(\omega t + \theta_v) \quad \text{and} \quad i(t) = I_m \cos(\omega t + \theta_i)$$

in phasor form, $\mathbf{V} = V_m \underline{/\theta_v}$ and $\mathbf{I} = I_m \underline{/\theta_i}$,

the average power is $P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$

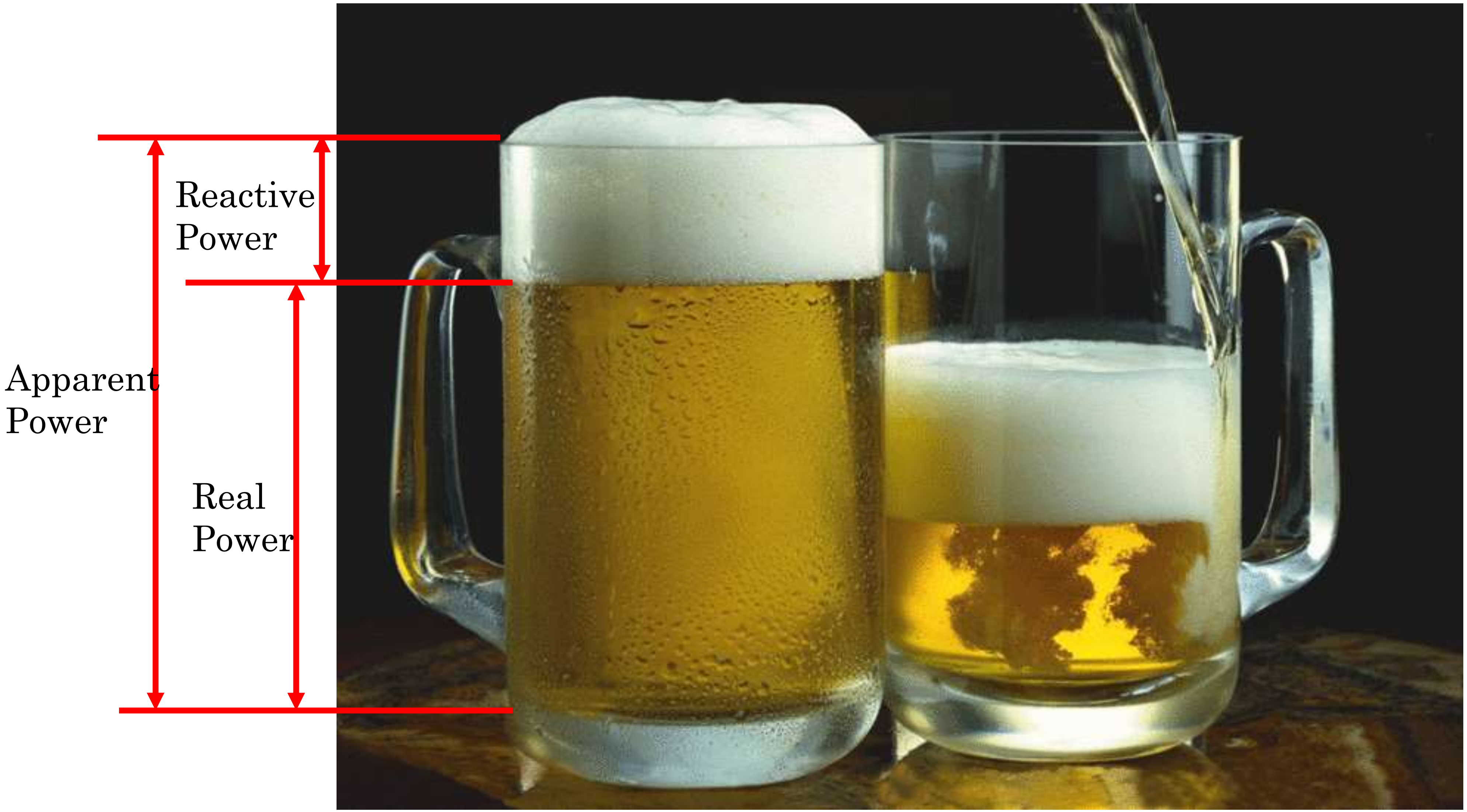
$$P = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) = S \cos(\theta_v - \theta_i)$$

Apparent power $\boxed{S = V_{\text{rms}} I_{\text{rms}}}$ VA

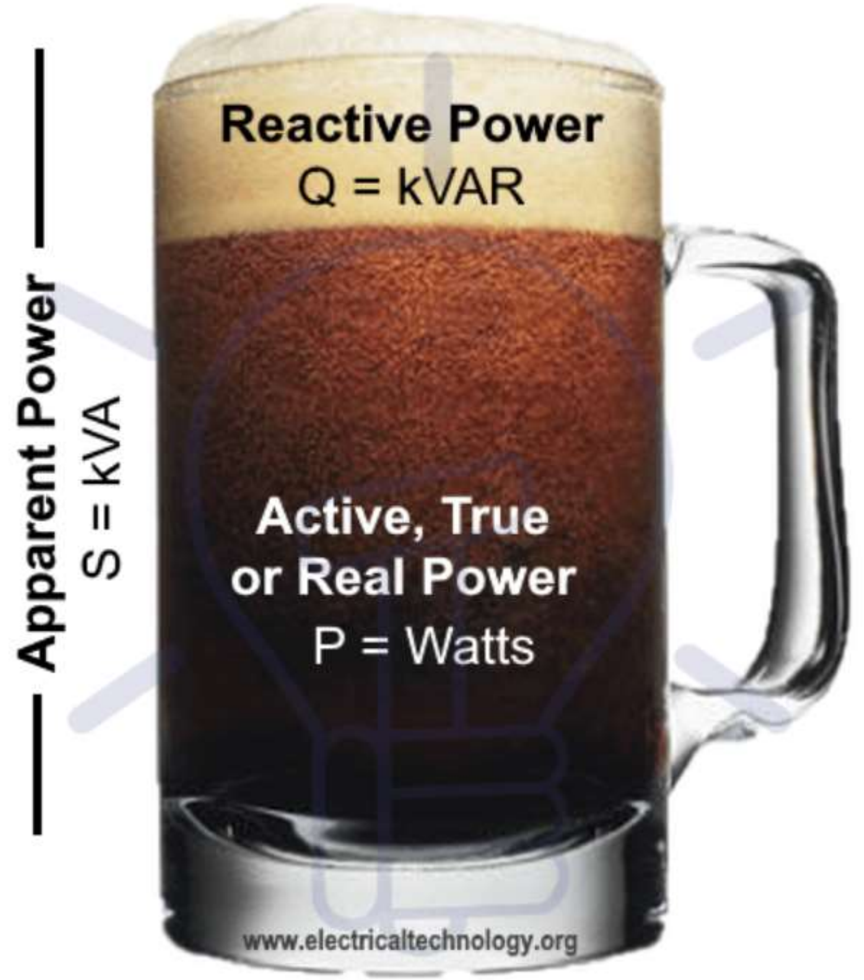
- The apparent power (in VA) is the product of the rms values of voltage and current.

$$\text{Apparent Power} = S = |\mathbf{S}| = |\mathbf{V}_{\text{rms}}|\,|\mathbf{I}_{\text{rms}}| = \sqrt{P^2 + Q^2}$$

Power factor = $\text{pf} = \frac{P}{S} = \cos(\theta_v - \theta_i)$


Reactive Power
Apparent Power
Real Power

Analogy of Active, Reactive and Apparent Powers



Analogy of Active, Reactive & Apparent Power

- **The real power *P*** is the rms/effective power in watts delivered to a load; it is the only useful power. It is the actual power dissipated by the load.

- **The reactive power *Q*** is a measure of the energy exchange between the source and the reactive part of the load. The unit of Q is the *volt-ampere reactive* (VAR) to distinguish it from the real power, whose unit is the watt.

- **The apparent power** is so called because it seems apparent that the power should be the voltage-current product, by analogy with dc resistive circuits. It is measured in volt-amperes or VA

- The real power P is the average power in watts delivered to a load; it is the only useful power. It is the actual power dissipated by the load.
- The reactive power Q is a measure of the energy exchange between the source and the reactive part of the load. The unit of Q is the volt-ampere reactive (VAR) to distinguish it from the real power, whose unit is the watt.
- We know that energy storage elements neither dissipate nor supply power, but exchange power back and forth with the rest of the network.
- In the same way, the reactive power is being transferred back and forth between the load and the source. It represents a lossless interchange between the load and the source.

1. $Q = 0$ for resistive loads (unity pf).
2. $Q < 0$ for capacitive load (leading pf)
3. $Q > 0$ for inductive load (lagging pf)

Overview:

$$\text{Complex Power} = \mathbf{S} = P + jQ = \mathbf{V}_{\text{rms}}(\mathbf{I}_{\text{rms}})^*$$

$$= |\mathbf{V}_{\text{rms}}|\,|\mathbf{I}_{\text{rms}}|\underline{/\,\theta_v - \theta_i}$$

$$\text{Apparent Power} = S = |\mathbf{S}| = |\mathbf{V}_{\text{rms}}|\,|\mathbf{I}_{\text{rms}}| = \sqrt{P^2 + Q^2}$$

$$\text{Real Power} = P = \text{Re}(\mathbf{S}) = S\cos(\theta_v - \theta_i)$$

$$\text{Reactive Power} = Q = \text{Im}(\mathbf{S}) = S\sin(\theta_v - \theta_i)$$

$$\text{Power Factor} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

1) The voltage across a load is $\boldsymbol{v(t) = 60cos(\omega t - 10°)V}$ and the current through the element in the direction of the voltage drop is $\boldsymbol{i(t) = 1.5cos(\omega t + 50°)A}$ Find: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

(a) For the rms values of the voltage and current, we write

$$\mathbf{V}_{\text{rms}} = \frac{60}{\sqrt{2}}\angle -10°, \qquad \mathbf{I}_{\text{rms}} = \frac{1.5}{\sqrt{2}}\angle +50°$$

The complex power is

$$\mathbf{S} = \mathbf{V}_{\text{rms}}\mathbf{I}^*_{\text{rms}} = \left(\frac{60}{\sqrt{2}}\angle -10°\right)\left(\frac{1.5}{\sqrt{2}}\angle -50°\right) = 45\angle -60° \text{ VA}$$

The apparent power is

$$S = |\mathbf{S}| = 45 \text{ VA}$$

(b) We can express the complex power in rectangular form as

$$\mathbf{S} = 45\angle -60^\circ = 45[\cos(-60^\circ) + j\sin(-60^\circ)] = 22.5 - j38.97$$

Since $\mathbf{S} = P + jQ$, the real power is

$$P = 22.5 \text{ W}$$

while the reactive power is

$$Q = -38.97 \text{ VAR}$$

(c) The power factor is

$$\text{pf} = \cos(-60^\circ) = 0.5 \text{ (leading)}$$

It is leading, because the reactive power is negative. The load impedance is

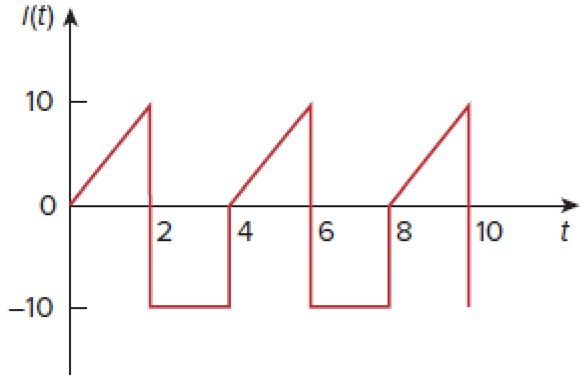
$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{60\angle -10^\circ}{1.5\angle +50^\circ} = 40\angle -60^\circ \ \Omega$$

which is a capacitive impedance.

7) Determine the rms value of the current waveform in Fig given. If the current is passed through a 2-Ω resistor, find the average power absorbed by the resistor.

- The period of the waveform is $T = 4$. Over a period, we can write the current waveform as

$$i(t) = \begin{cases} 5t, & 0 < t < 2 \\ -10, & 2 < t < 4 \end{cases}$$



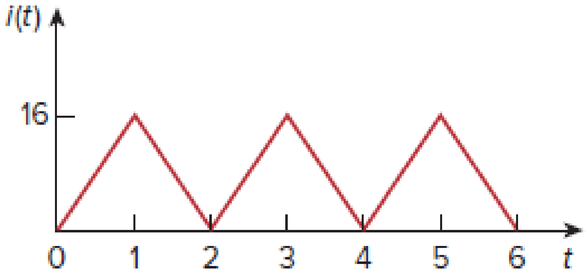
- The rms value is

$$I_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T i^2\,dt} = \sqrt{\frac{1}{4}\left[\int_0^2 (5t)^2\,dt + \int_2^4 (-10)^2\,dt\right]}$$

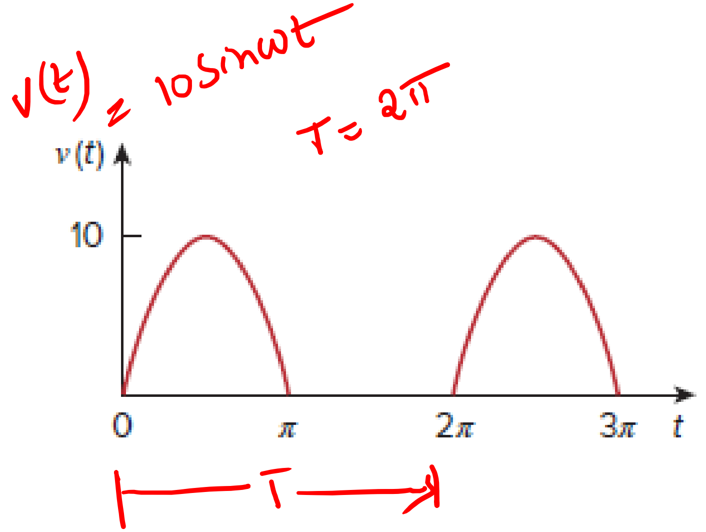
$$= \sqrt{\frac{1}{4}\left[25\frac{t^3}{3}\Big|_0^2 + 100t\Big|_2^4\right]} = \sqrt{\frac{1}{4}\left(\frac{200}{3} + 200\right)} = 8.165\ \text{A}$$

- The power absorbed by a 2-Ω resistor is $P = I_{\text{rms}}^2 R = (8.165)^2(2) = 133.3\ \text{W}$

8) Find the rms value of the current waveform of Fig below. If the current flows through a 9-Ω resistor, calculate the average power absorbed by the resistor.



9) The waveform shown in Fig. below is a half-wave rectified sine wave. Find the rms value and the amount of average power dissipated in a 10-Ω resistor.



- Find the rms value of the full-wave rectified sine wave in Fig. below. Calculate the average power dissipated in a 6-Ω resistor.

$v(t) = 100 \sin \omega t$

$T = \pi$